RM500U Series

Reference Design

5G Module Series

Version: 1.2

Date: 2023-12-20

Status: Released

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About the Document

Revision History

Version	Date	Author	Description
-	2022-12-09	Len CHEN	Creation of the document
1.0	2023-02-27	Len CHEN	First official release
1.1	2023-08-31	Len CHEN	Added the applicable modules RM500U-EA and RM500U-CNV.
1.2	2023-12-20	Len CHEN	Added a note on avoiding abnormal RF functions caused by current sink on the module's pins (Sheet 1).

Contents

1 Reference Design

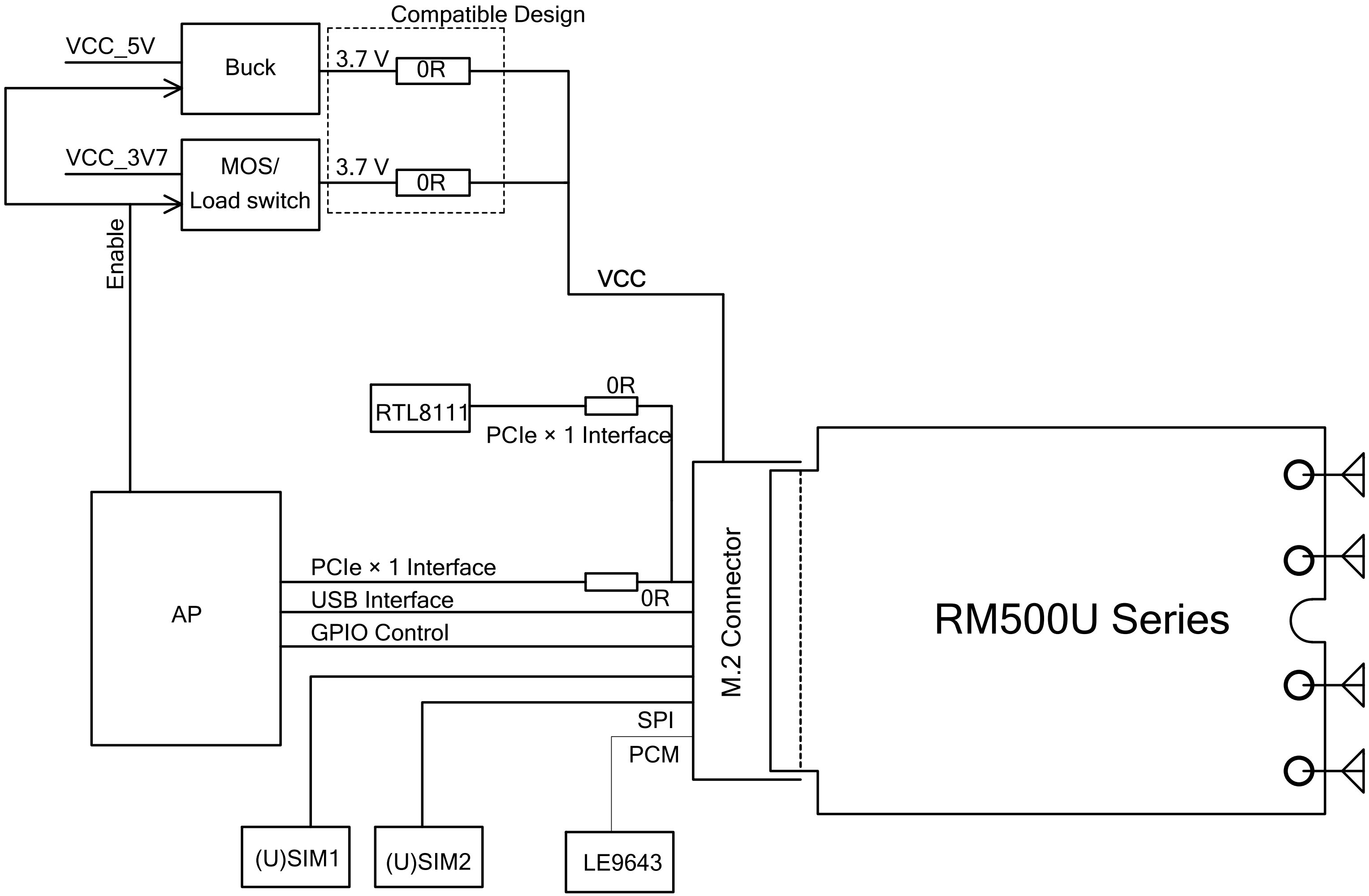
1.1. Introduction

This document provides the reference design of Quectel RM500U series module. The reference design includes block diagram, power supply designs, module interfaces, AP interfaces, (U)SIM interface design, Ethernet and SLIC solution, etc.

1.2. Schematics

The schematics illustrated in the following pages are provided for your reference only.

Block Diagram



NOTE:

Ensure that the pull-up power supply of the module's pins is VDD_EXT or controlled by VDD_EXT, and there is no current sink on the module's pins before the module turns on.

For more details, contact Quectel Technical Support.

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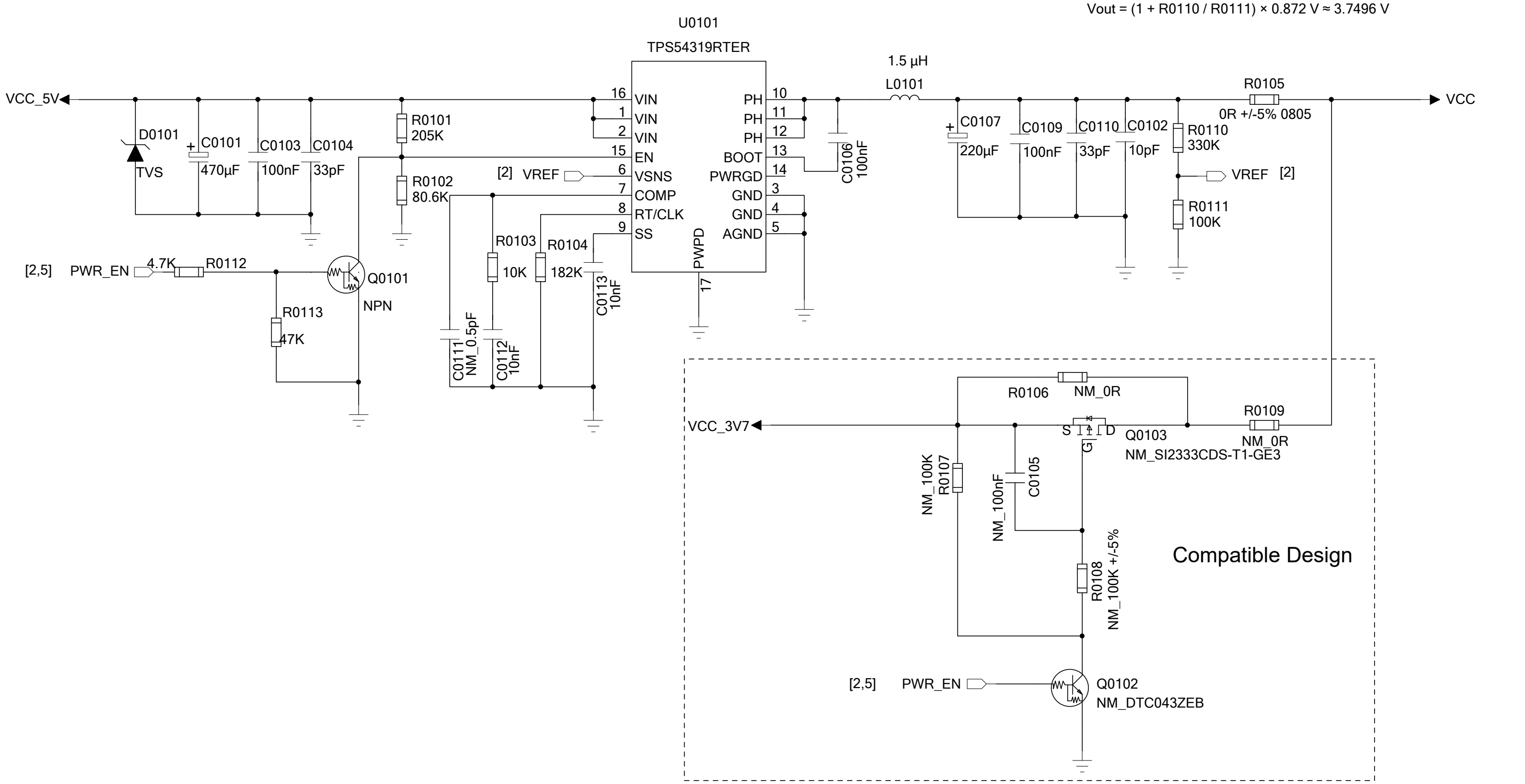
Power Supply Design (Part 1)

Lmin = (Vin - Vout) × Vout / (0.4 × Iout(max) × Fsw × Vin) = (5.0 V - 3.7 V) × 3.7 V / (0.4 × 3.0 A × 1.0 MHz × 5.0 V) ≈ 0.802 µH

Vout = (1 + R0110 / R0111) × 0.872 V ≈ 3.7496 V



NOTE:

The power supply is critical to the module's performance. The continuous current of the power supply should be 3 A at least and the peak current should be 4 A at least.

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Power Supply Design (Part 2)

SLIC Power Supply

Vout = (1 + R0202 / R0203) × 1.207 V ≈ 3.312 V

Vout = (1 + R0205 / R0206) × 1.207 V ≈ 1.835 V

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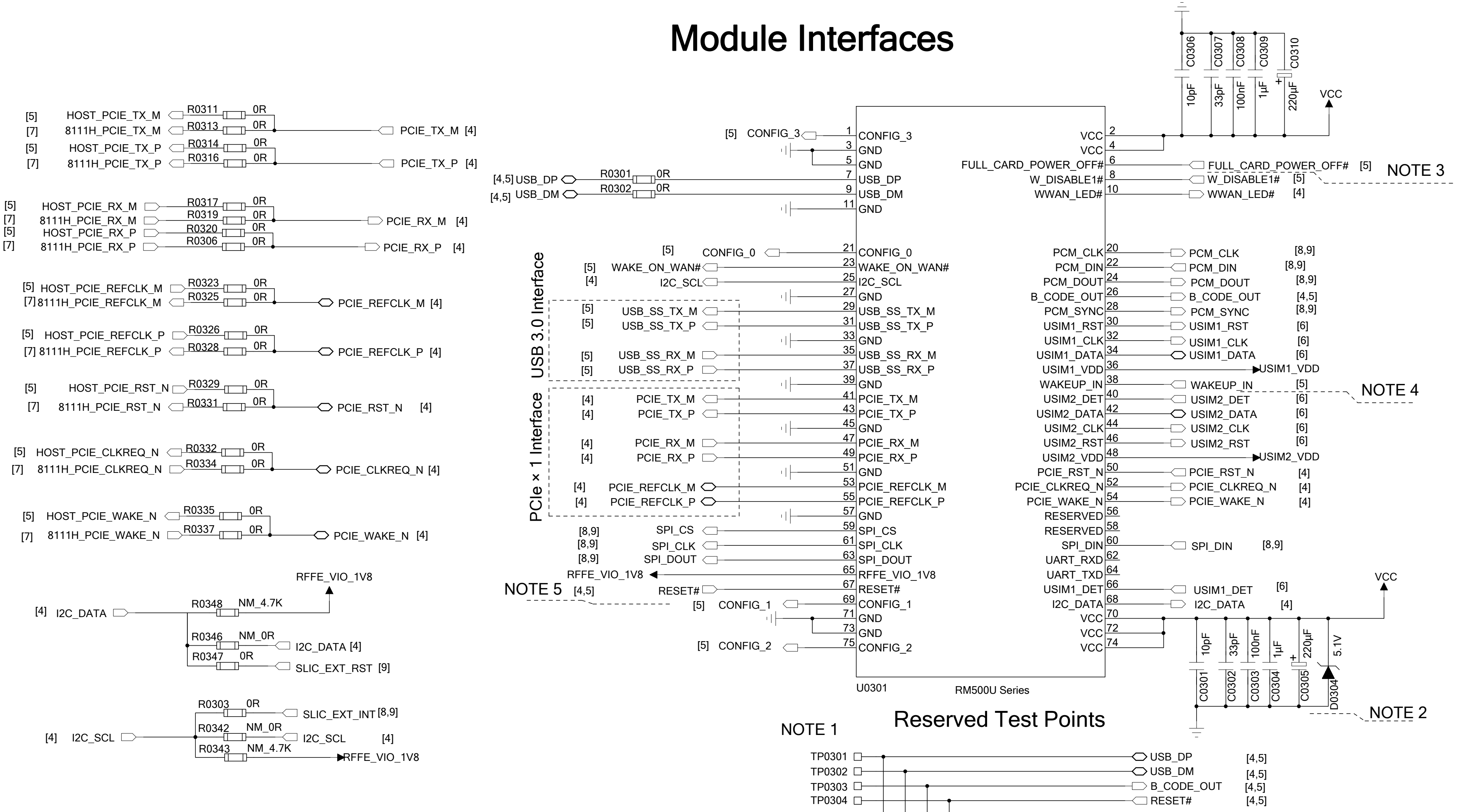
Module Interfaces



NOTE:

1. Test points must be reserved for USB 2.0 interface for firmware upgrade, and the stub length of USB test points should be minimized.
   Ensure that the stray capacitance of the ESD protection components is less than 1 pF. B_CODE_OUT and RESET# are recommended to be reserved if unused.
2. It is recommended to add a voltage regulator near the module power supply pins.
3. Use the GPIO interface of the AP to control the FULL_CARD_POWER_OFF# of the module.
4. WAKEUP_IN is used for sleep mode control of the module, and it is internally pulled up to 1.8 V with a 100 kΩ resistor. When it is driven low, the module enters sleep mode,
   and when it is driven high, the module can be woken up from sleep mode.
5. RESET# is internally pulled up with a 20 kΩ resistor to VCC and cannot be directly connected to the GPIO of the host. A NPN driver circuit, NMOS driver circuit or a button can be used to control the pin.
   And a test point is recommended to be reserved if unused.
6. RFFE_VIO_1V8 is the 1.8 V power supply for external pull-up circuits. If the signal that connects to the module needs to be pulled up to 1.8 V, you must select the RFFE_VIO_1V8 as the pull-up power supply.

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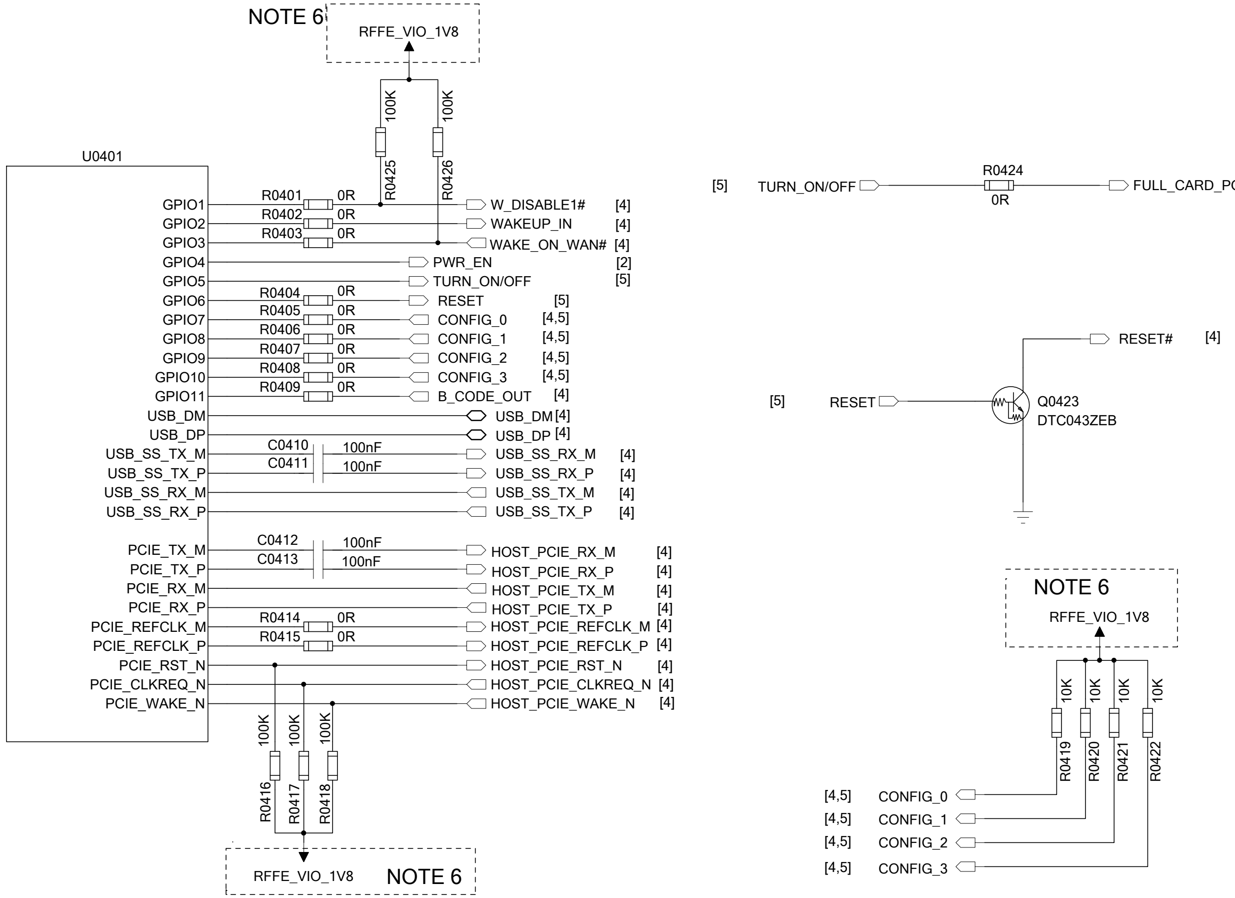
AP Interfaces



NOTE:

1. U0401 represents your device.
2. Ensure that C0410-C0413 are placed as close to your device as possible.
3. The impedance of the USB 2.0 and USB 3.0 signal traces needs to be controlled at 90 Ω.
4. The impedance of the PCIe signals needs to be controlled at 100 Ω ±10 %.
5. Keep the ESD protection components as close to the USB connector as possible.
6. For an AP with 1.8 V power domain, RFFE_VIO_1V8 can be used as the pull-up power supply.
   For an AP with 3.3 V power domain, RFFE_VIO_1V8 should control whether to enable the 3.3 V power supply.

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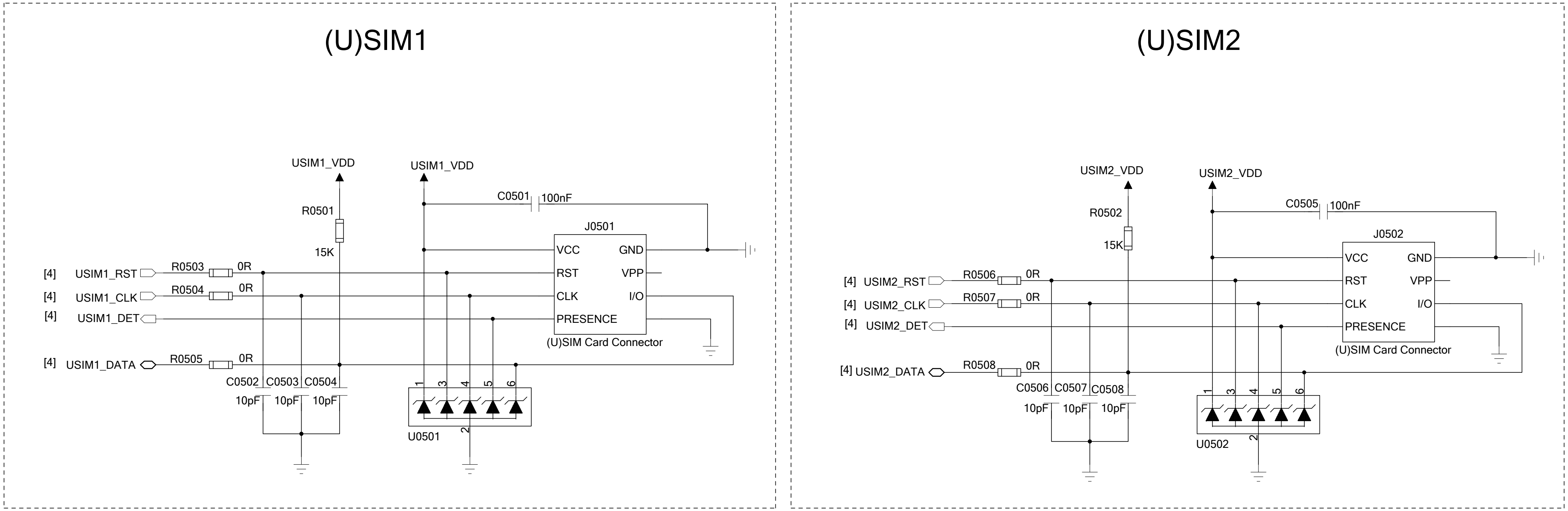
(U)SIM Interface Design



NOTE:

1. The decoupling capacitors of USIM1_VDD and USIM2_VDD should be placed close to the (U)SIM card connector.
2. The module provides USIM1_DET and USIM2_DET for detecting (U)SIM cards.
   This reference design is only for normally closed (U)SIM card connector that supports high and low level detection.
3. The 0 Ω resistors (R0503-R0508) should be added in series between the module and the (U)SIM card connector to suppress EMI spurious transmission and enhance ESD protection.
4. To offer better ESD protection, it is recommended to add a TVS array (U0501 and U0502) of which a parasitic capacitance not exceeding 10 pF, and place it as close as possible to the (U)SIM card conneor.
5. The (U)SIM card connector needs to be placed close to the M.2 connector as long PCB traces will affect the signal quality.

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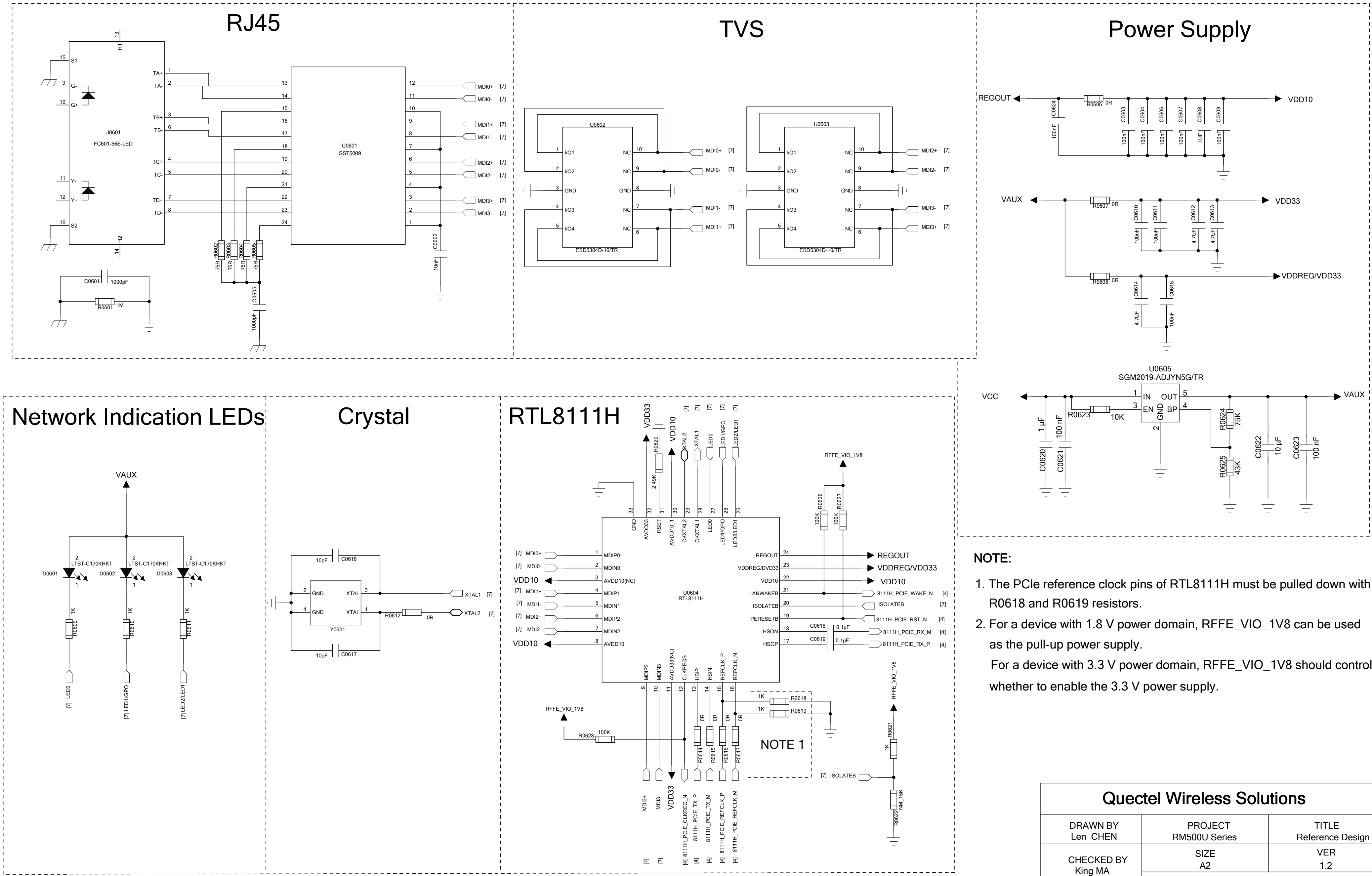
Ethernet



NOTE:

1. The PCIe reference clock pins of RTL8111H must be pulled down with R0618 and R0619 resistors.
2. For a device with 1.8 V power domain, RFFE_VIO_1V8 can be used as the pull-up power supply.
   For a device with 3.3 V power domain, RFFE_VIO_1V8 should control whether to enable the 3.3 V power supply.

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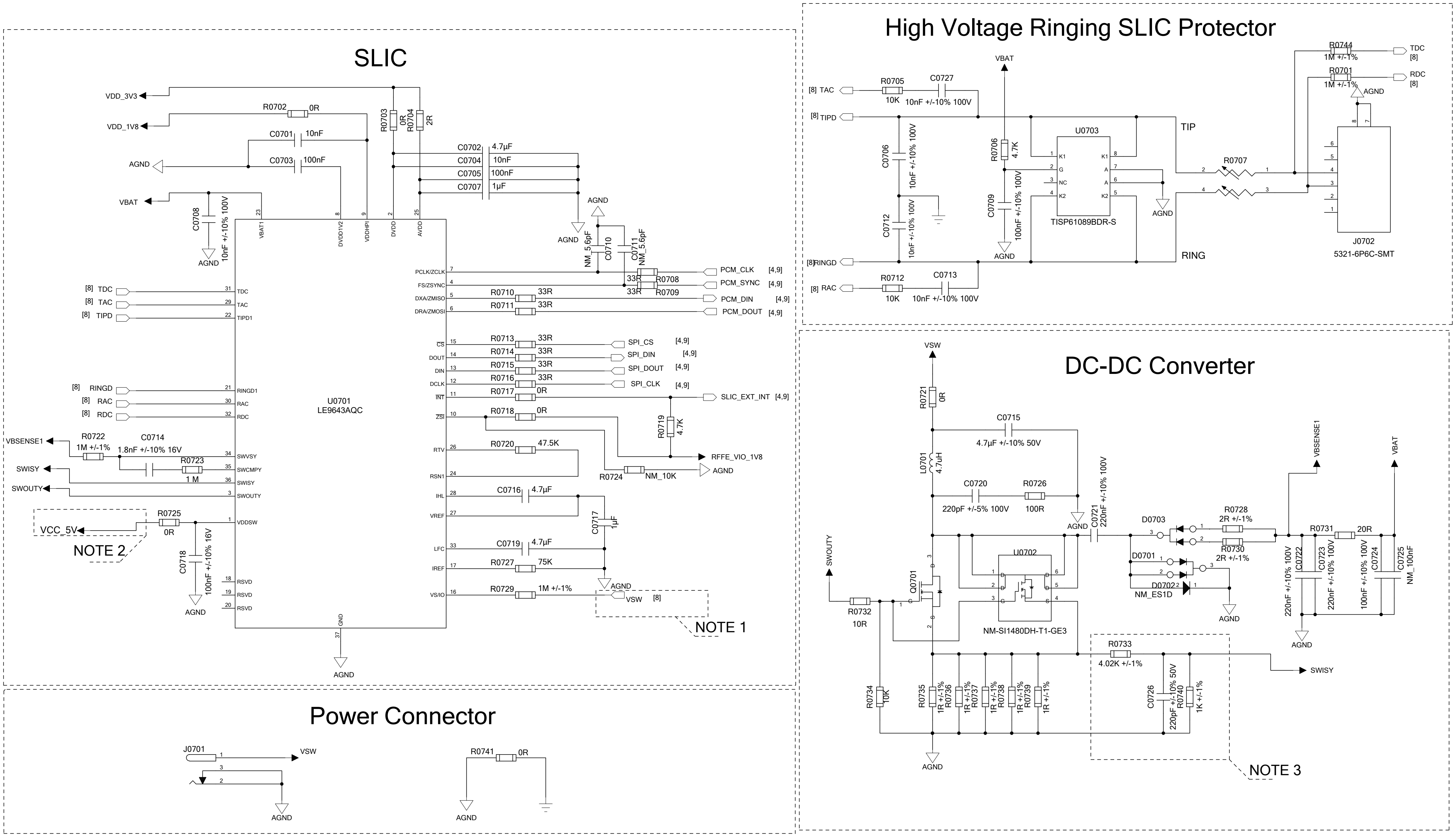
SLIC Solution 1



NOTE:

1. The VSW input voltage range is 10-14 V.
2. VDDSW voltage is +5 V ±5 %, and the absolute maximum voltage is +5.5 V.
3. The current detection circuit needs to be placed close to the SLIC chip.
4. It is required to confirm the applicability and price from the supplier about the IC involved in the reference design.

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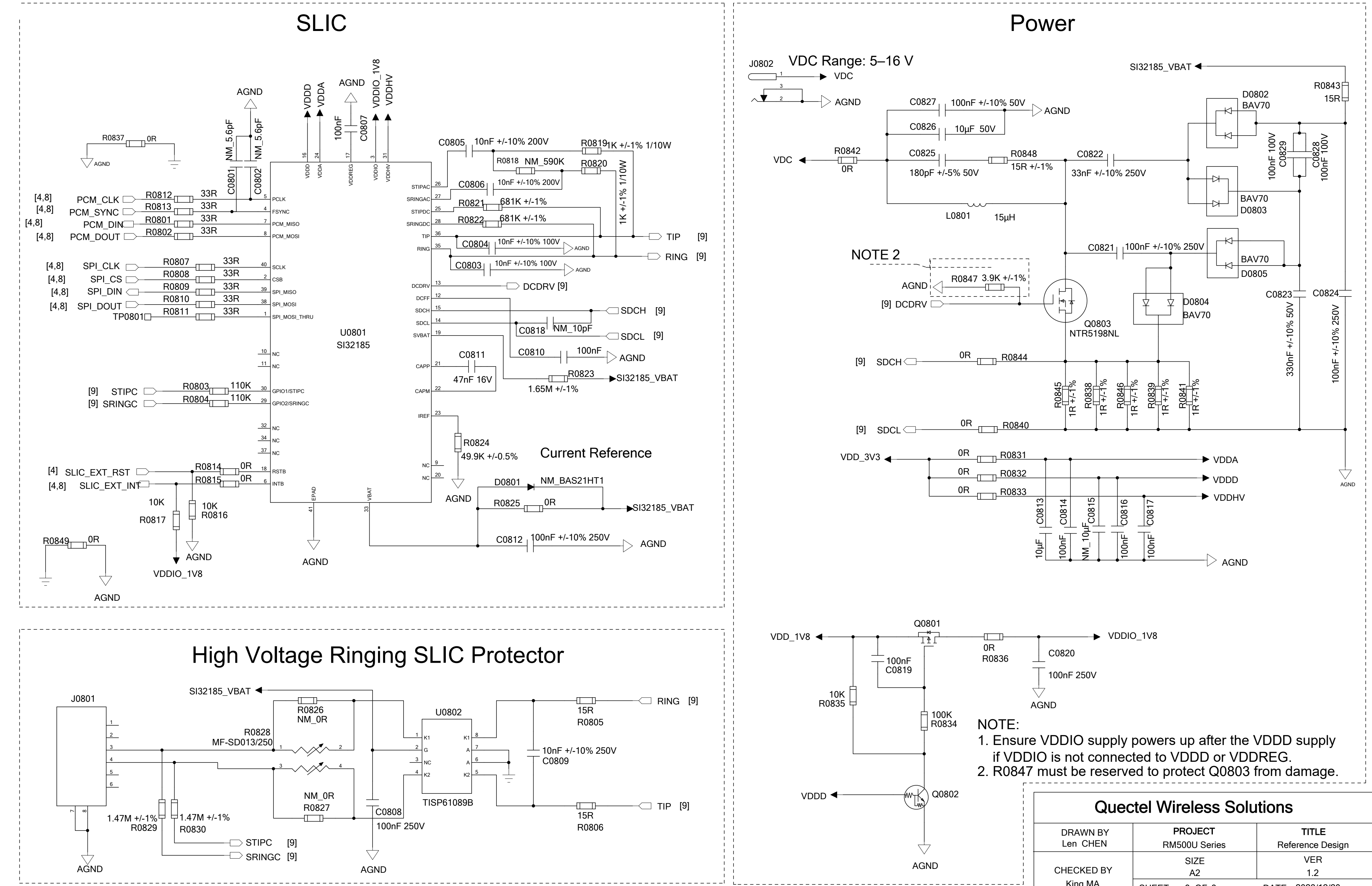
SLIC Solution 2

SLIC

Power

High Voltage Ringing SLIC Protector



NOTE:
1. Ensure VDDIO supply powers up after the VDDD supply if VDDIO is not connected to VDDD or VDDREG.
2. R0847 must be reserved to protect Q0803 from damage.

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